

MGT 7316 Module Agendas

Module 1: Requirements Engineering (Week 1 and Week 2)

Overview

The module will expose learners to the importance, process, and features of a product requirements document (PRD). The purpose of the module is to allow learners to develop skills in defining clear and concise functional and non-functional requirements, implementing scope management, decision-making, and composing formal technical writing documentation.

Learning Objectives

At the end of this module, you should be able to:

- Develop a PRD overview and list goals and objectives.
- Generate user stories based on the
- Define functional (e.g., features) and non-functional (e.g., performance, quality, security, accessibility) requirements.
- Define user interface or visual style guidelines
- Define the definition of done (aka acceptance criteria)
- Document all assumptions, axioms, or constraints
- Document all internal or external dependencies
- Document timelines (e.g., effort vs. duration vs. elapsed)

Learning Activities

Web Tools

Replit integrated development environment
Google Chrome Browser
LMS assignment submission and discussion boards

Lectures

Course Introduction Slides
Wicked Business Problems Slides
Defining “users”, “requirements”, and “done” Slides

Readings

[Product Requirements Docs \(Atlassian\)](#)
[Google Technical Writing One](#)
[Markdown Guide](#)

Discussion Board

After completing the assigned readings, reflect and answer the following questions: What is the importance of creating a PRD? What are the implications of poorly defined requirements? Visit [Google Cloud \(https://cloud.google.com/\)](https://cloud.google.com/) and, based on the verbiage and terminology on the landing page and supporting (linked) pages, assert five functional requirements Google may have defined for the Google Cloud Platform. Post a new thread post and two replies to your peers by the end of Week 2.

Evaluation Activities

Module 1 Assignment: Product Requirement Document (PRD)

You have been hired as a software engineer at a fictional software company called Ctrl-Alt-Delight. Create a comprehensive Product Requirements Document (PRD), using Markdown syntax, for a new consumer-facing web application called Learning Hurts, which helps adult learners explore new topics and materials effectively. The PRD will serve as a blueprint for the development team, outlining the application's users, functionalities, and specifications. Include the following sections in your PRD:

- Product description, including the product purpose and intended users.
- Product scope, including functional features
- User interface and style guidelines and conformity
- User stories using Gherkin format
- Non-functional requirements (e.g., web platform, software, security)
- Technical dependencies, including integration, data management
- Acceptance Criteria (definition of done)
- Estimated Timelines
- (Optional) Appendices

Create a Markdown file named README.md placed at the root level of a Replit repo and submit the Replit repo URL into the LMS.

Pre-Assessment Quiz

Learners will take an assessment before Module 1 to assert their prior knowledge and skill regarding programming techniques, business writing, and composition. The assessment intends to identify any addressable misconceptions or gaps and provide individualized feedback and tracking as the course progresses.

Question 1

On a scale from 1 to 10, how do you estimate your programming experience?

Question 2

How would you estimate your programming experience compared to your peers?

Question 3

How many years have you been programming?

Question 4

How many years have you been programming for larger software projects, if any?

Question 5

What is the MVC pattern?

Question 6

What is the Test Pyramid?

Question 7

Can you interpret the intent of the following file?

```
# cloudbuild.yaml
steps:
- name: 'gcr.io/cloud-builders/gcloud'
  entrypoint: 'bash'
  args:
    - '-c'
    - |
      gcloud functions deploy <YOUR_FUNCTION_NAME> \
        --runtime <YOUR_PREFERRED_RUNTIME> \
        --trigger-http \
        --allow-unauthenticated \
        --region <YOUR_REGION> \
        --project <YOUR_PROJECT_ID>
- name: 'gcr.io/cloud-builders/gcloud'
  entrypoint: 'bash'
  args:
    - '-c'
    - |
      gsutil mb -p <YOUR_PROJECT_ID> -l <YOUR_BUCKET_LOCATION> gs://<YOUR_BUCKET_NAME>
- name: 'gcr.io/cloud-builders/gcloud'
  entrypoint: 'bash'
  args:
    - '-c'
    - |
      gcloud firestore databases create --region <YOUR_FIRESTORE_REGION> --project <YOUR_PROJECT_ID>
```

Question 8

Describe the parts of a product requirements document.

Question 9

Describe the parts of a solution architecture and application design document.

Question 10

Describe Agile, Scrum, and Kanban.

Module 2: Web Application Architecture and Design (Week 3 and Week 4)

Overview

The module will expose learners to commonly used web application architectures and design patterns. The purpose of the module is to allow learners to conceptualize the client-server model and account for performance and maintenance considerations in their proposed solution architectures.

Learning Objectives

At the end of this module, you should be able to:

- Define a web application architecture using a combination of technical writing and architectural diagrams.
- Document and characterize architectural details such as scaling, interoperability, and maintenance.
- Compose system, component, and user interaction diagrams to accompany written documentation
- Outline data structures, integrations, and technology stacks and layers

Learning Activities

Web Tools

Replit integrated development environment
Google Chrome Browser
LMS assignment submission and discussion boards

Lectures

Model-View-Controller Slides
Client-Server Model Slides
Microservice Pattern Slides

Readings

Docs for Devs (Chapter. 6)
[Software Architecture Guide](#)
[Model-View-Controller \(Mozilla Developer Network\)](#)

Discussion Board

After completing the assigned readings, reflect and answer the following questions:
What is the importance and purpose of creating a SADD in the product development lifecycle?
Visit [Google Drive \(https://www.google.com/drive/\)](https://www.google.com/drive/) and, based on the verbiage and terminology on the landing page and supporting (linked) pages, assert five functional requirements Google may have defined for Google Drive. Then, create a high-level solution architecture for how Google Drive could work. Post a new thread post and two replies to your peers by the end of Week 2.

Evaluation Activities

Module 2 Assignment: Web Application Solution Architecture and Design Document (SADD)

You're (still) a software engineer at a fictional software company called Ctrl-Alt-Delight. Create a solution architecture and design document (SADD), using Markdown syntax, for a new consumer-facing web application called Learning Hurts, which helps adult learners explore new topics and materials effectively. The SADD will serve as a guide for developing the Learning Hurts web application and implementing the functional and non-functional requirements of the Module 1 PRD while meeting /exceeding the acceptance criteria. Include the following sections in your SADD:

- Architectural description, including a brief of the product objectives
- Functional scope, architectural goals, and design objectives (e.g., interoperability, compatibility, go to market)
- Non-functional requirements
- Architectural diagrams, including systems, interactions, and integrations)
- Data Architecture, including models, flow, storage, and lifecycle management
- Technology stack and layers
- Risks and challenges
- Future enhancements, maintenance needs
- (Optional) Appendices

Create a Markdown file named SADD.md placed at the root level of the Replit repo used for Module 1 README.md and submit the Replit repo URL into the LMS.

Module 3: Web Application Programming and Development (Week 5 and Week 6)

Overview

The module will allow learners to implement several functional features of a web application using established programming design patterns. The purpose of the module is to expose learners to gaps between architectural proposals and implementation.

Learning Objectives

At the end of this module, you should be able to:

- Implement web component definitions and functional modules
- Implement data models and schemas
- Define integration points, patterns, and web API contracts
- List software development tools/apps and package dependencies (including semantic versions)
- Implement security and privacy measures (e.g., cryptographic mechanisms)

Learning Activities

Web Tools

Replit integrated development environment
Google Chrome Browser
LMS assignment submission and discussion boards

Lectures

Web Programming Pattern Slides
Serverless Slides
Module Pattern Slides

Readings

[Object Oriented Programming \(Mozilla Developer Network\)](#)
[JavaScript Patterns](#)
[Google Cloud Functions](#)
[Google Cloud Storage: Firebase](#)
[Google Cloud Storage: Buckets](#)

Discussion Board

After completing the assigned readings, reflect and answer the following questions:
What parts of the SADD implementation were missing or undefined during the implementation of the Learning Hurts web application? What parts of the SADD needed to be updated or changed during the implementation of the Learning Hurts web application? Post a new thread post and two replies to your peers by the end of Week 2.

Evaluation Activities

Module 3 Assignment: Learning Hurts Web Application Development

You're (still) a software engineer at a fictional software company called Ctrl-Alt-Delight. Code/Implement the functional features of the Learning Hurts web application specified in the Module 2 SADD with the following considerations:

- Minimize repetitious code / Maximize reusable code (e.g., modules)
- Practice well-developed naming conventions for functions and variables/constants
- Organize code into subfolders by function
- Practice [separation of concerns](#) and [single responsibility principle](#)

Create a folder named `src` at the root level of the Replit repo used for Module 1 README.md and Module 2 SADD.md containing the source code for your web application implementation. Submit the Replit repo URL into the LMS.

Module 4: Web Application Deployment (Week 7 and Week 8)

Overview

The module will expose learners to testing, refactoring, and deploying a functional web application to a production-based cloud service. The purpose of the module is to allow learners to walk through the steps of testing and validating functionality, updating, and packaging application code, and installing an app on (serverless) cloud resources.

Learning Objectives

At the end of this module, you should be able to:

- Define unit tests for functions and modules of code.
- Define functional (end-to-end) tests for each application feature.
- Define integration test suite for all features
- Define regression tests for future enhancements and extensions
- Complete web application code packaging and deployment

Learning Activities

Web Tools

Replit integrated development environment
Google Chrome Browser
LMS assignment submission and discussion boards

Lectures

Cloud Deployment Pattern Slides
Unit/Functional Testing
Refactoring

Readings

[Practical Test Pyramid](#)
[Google Cloud Deploy](#)
[Refactoring](#)

Discussion Board

After completing the assigned readings, reflect and answer the following questions:
What are the differences between unit, functional, and regression tests? What is a “test double”, and provide two examples. Where can mocks, stubs, and spies be used in unit testing? Describe how what a deployment pipeline is and provide an example of how the Learning Hurts application deployment could be accomplished via a Google Cloud Deploy pipeline. Post a new thread post and two replies to your peers by the end of Week 2.

Evaluation Activities

Module 4 Assignment: Web Application Solution Testing and Deployment

You're (still) a software engineer at a fictional software company called Ctrl-Alt-Delight. Compose unit tests for each function/module written for the Learning Hurts web application created in Module 3. Each test should cover at least two use cases (one positive, one negative). In addition, using two or more unit tests, one functional test suite should be written to test each functional feature with at least two use cases (one positive, one negative).

Create a folder named `tests` at the root level of the Replit repo used for Module 1 README.md, Module 2 SADD.md, and Module 3 `src` folder containing the test code for your web application. Submit the Replit repo URL into the LMS.

Module 5: Team Project (Week 9 and Week 10)

Overview

The module will expose collaborate in a two-week Agile sprint to design, develop, and deploy a web application of their choice. The purpose of the module is to provide practical experience with Agile, Scrum, and Kanban rituals and practices for managing the product development lifecycle.

Learning Objectives

At the end of this module, you should be able to:

- Working groups can define and generate product requirements documentation (PRD)
- Working groups can define a feasible web application solution architecture design document (SADD) matching the previously developed PRD.
- Working groups develop a web application based on web application solution architecture and design document (SADD)
- Working groups can implement unit and functional test suites for their web application source code and develop deployment pipeline files.

Learning Activities

Web Tools

Replit integrated development environment
Google Chrome Browser
LMS assignment submission and discussion boards

Lectures

Agile Project Management Slides

Readings

[Agile Project Management](#)
[Kanban](#)
[Scrum](#)

Discussion Board

After completing the assigned readings, reflect and answer the following questions:
What is Agile? How does Agile compare to waterfall processes? What are the benefits of Scrum and Kanban? Post a new thread post and two replies to your peers by the end of Week 2.

Evaluation Activities

Module 5 Assignment: Team Project

As a working group, you have been hired as software engineers at a fictional software company

called Ctrl-Alt-Delight. Ideate and develop a product concept for a software application that would be helpful for undergraduate and graduate students to improve their learning speed. Create a comprehensive Product Requirements Document (PRD) and Solution Architecture and Design Document (SADD) using Markdown syntax, compose source code for a web application that implements the specifications of the PRD and SADD, compose unit and functional tests that validate and verify the web application, and compose a deployment pipeline to deliver the application to Google Cloud Platform.

Create a new team-accessible Replit repo, write and place your PRD and SADD Markdown files at the root folder level, write and place your source code in a folder named **src**, write and place your test code in a folder named **tests**, deploy the web application to Google Cloud Platform and submit Replit repo URL AND deployment URL into the LMS.

Post-Assessment Quiz

Learners will take an assessment following Module 5 to measure learner progress and identify any learning or skill gaps between the desired and actual learning goals.

Question 1

On a scale from 1 to 10, how do you estimate your programming experience?

Question 2

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Question 4

How many years have you been programming for larger software projects, if any?

Question 5

What is the MVC pattern?

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What is the Test Pyramid?

Question 7

What is the intent of the following file?

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# cloudbuild.yaml
steps:
- name: 'gcr.io/cloud-builders/gcloud'
  entrypoint: 'bash'
  args:
  - '-c'
  - |
    gcloud functions deploy <YOUR_FUNCTION_NAME> \
      --runtime <YOUR_PREFERRED_RUNTIME> \
      --trigger-http \
      --allow-unauthenticated \
      --region <YOUR_REGION> \
      --project <YOUR_PROJECT_ID>
- name: 'gcr.io/cloud-builders/gcloud'
  entrypoint: 'bash'
  args:
  - '-c'
  - |
    gsutil mb -p <YOUR_PROJECT_ID> -l <YOUR_BUCKET_LOCATION> gs://<YOUR_BUCKET_NAME>
```

```
- name: 'gcr.io/cloud-builders/gcloud'
  entrypoint: 'bash'
  args:
    - '-c'
    - |
      gcloud firestore databases create --region <YOUR_FIRESTORE_REGION> --project <YOUR_PROJECT_ID>
```

Question 8

Describe the parts of a product requirements document.

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